

$-2 \Delta \ln L$

**CMS**  
*Internal*

$r = 3.783^{+0.717}_{-1.672}$

$r = 3.783^{+0.216}_{-0.270}$  (theory)  $^{+0.000}_{-0.000}$  (TTnorm)  $^{+0.285}_{-0.212}$  (PF)  $^{+0.136}_{-0.111}$  (lumi)  $^{+0.045}_{-0.034}$  (btag)  $^{+0.189}_{-0.030}$  (jet)  $^{+0.365}_{-0.321}$  (Pileup)  $^{+0.068}_{-0.035}$  (VBS)  $^{+0.034}_{-0.024}$  (MET)  $^{+0.227}_{-0.195}$  (tau)  $^{+0.044}_{-0.000}$  (lepton)  $^{+0.000}_{-0.000}$  (mischarge)  $^{+1.666}_{-1.088}$  (MCstat)  $^{+0.779}_{-1.294}$  (Stat)

- Total uncert.
- - - Freeze TTnorm
- - - Freeze lumi
- - - Freeze jet
- - - Freeze VBS
- - - Freeze tau
- - - Freeze mischarge
- - - Freeze theory
- - - Freeze PF
- - - Freeze btag
- - - Freeze Pileup
- - - Freeze MET
- - - Freeze lepton
- - - Freeze all

